IM1 Book 1 Selected Answers

IM1 Dream Team

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1. (a) $A_{ABCD} = 25,\ A_{BCEF} = 9$
   (b) –
   (c) –
   (d) $A = 34$
   (e) $l = \sqrt{34}$
   (f) –
2. $l = 4\sqrt{5}$
3. –
4. $AB = \sqrt{41}$
5. $l = 5\sqrt{2}$
6. $l = \sqrt{5},$ No
7. $d = 10\sqrt{2}$
8. 12
9. $(12, 2), (2, 2)$
10. No
11. (a)

$$SR = PQ = 13$$
$$RQ = PS = 17$$
$$m_{SR} = m_{PQ} = \frac{-5}{12}$$
$$m_{RQ} = m_{PS} = \frac{15}{8}$$

   (b) It is a parallelogram but not a rectangle.
12. $(-6, 3), (10, 3)$

13. $AB = BC = \sqrt{10}$

14. (a) $(5, 0)$ – Answers may vary.
    (b) $(5, 1)$ – Answers may vary.
    (c) $x = 5$
    (d) –

15. (a) $AP = BP = 2\sqrt{5}$
    (b) $(2, 2), (3, 5), (4, 8)$
    (c) No
    (d) $y = 3(x - 1) - 1$

16. (a) $(2, 5), (3, 4)$ – Answers may vary.
    (b) $(1, 4), (2, 3)$ – Answers may vary.

17. $(1, 5)$ – Answers may vary.
    There are infinite options.

18. (a) Same size, different shape, incomparable orientations – the pieces aren't equal.
    (b) Same size, same shape, different orientations – the pieces are equal

    Definition: Two figures are *congruent* if they have the same size and shape.

19. $y = \frac{1}{3}(x - 2) + 2$

20. $(0, 5), (3, 4), (4, 3), (5, 0), (4, -3), (3, -4), (0, -5), (-3, -4), (-4, -3), (-5, 0), (-4, 3), (-3, 4)$

21. $(2, 3), (2, -3)$

22. (a) Answers may vary: $(0, 0), (2, 5)$.
    (b) No

23. Answers may vary: $(0, 0), (4, 1), (8, 0), (4, -1)$

24. $(0, 5 + 4\sqrt{2}), (0, 5 - 4\sqrt{2})$

25. 6

26. Leg 1: $24 - 12\sqrt{2}$
    Leg 2: $24 - 12\sqrt{2}$
    Hypotenuse: $24\sqrt{2} - 24$

27. Yes

28. (a) If point P is equidistant from the coordinate axes, then point P is on the line $y = x$.

(b) No

(c) Answers may vary. If quadrilateral $ABCD$ is a square, then quadrilateral $ABCD$ is a rectangle.

(d) Answers may vary. If quadrilateral ABC has four right angles, then quadrilateral $ABCD$ is a rectangle.

29. They have the same side lengths.

30. (a) $\triangle ABC$ must be the same size and shape as $\triangle DEF$.
If two triangles have the same three side lengths, then the two triangles are congruent.

(b) $\overline{KL}$

(c) Typo: The question should say "What angle of $\triangle ABC$ corresponds to $\angle LKM$?"
Answer: $\angle BAC$

(d) $\overline{GI} \cong \overline{GI}$, so there are three pairs of congruent segments.

(e) They are congruent.

31. –

32. $\angle ABC$

33. $\overline{AB}$

34. (a) $PNMRQ$

(b) $\angle Q$

35. They must add to 90°.

36. It must be 90°.

37. $d = \sqrt{(x-2)^2 + (y-4)^2}$

38. (a) –

(b) –

(c) –

(d)

$$m_{QR} = \frac{-a}{b}$$
$$m_{QP} = \frac{b}{a}$$

39. –

40. –

41. –

42. –

43. –

44.

$$\triangle LAG \cong \triangle LEB$$
$$\triangle LEG \cong \triangle LAB$$
$$\triangle AGB \cong \triangle EBG$$
$$\triangle AGE \cong \triangle EBA$$

45. –

46. (a) $y - 2 = \frac{1}{2}(x + 1)$
    (b) $y = \frac{4}{5}(x - 2)$
    (c) $y = -\frac{5}{4}(x - 2)$

47. (a) –
    (b) Q; C.P.C.T.C.

48. $n = \frac{49}{4}$

49. (a) $\overline{AC} = \overline{AB} + \overline{BC}$; segment addition
    (b) –

50. Yes

51. No

52. C.P.C.T.C.

53. (a) $\angle ABD + \angle DBC = \angle ABC$; angle addition
    (b) –

54. 18 minutes

55. The slope is the speed of the plane's descent.
    The y-intercept is the plane's location when initially measured.
    The x-intercept is the time when the plane lands.

56. $m = -1$

57. –

58. –

59. (a) $\sqrt{(x+1)^2 + (y-5)^2}$

(b) $\sqrt{(x-5)^2+(y-2)^2}$

(c) $\sqrt{(x+1)^2+(y-5)^2} = \sqrt{(x-5)^2+(y-2)^2}$

(d) $y = 2x - \frac{1}{2}$

(e) $(2, \frac{7}{2})$

(f) $m_{AB} = -\frac{1}{2}$

(g) –

60. All points (x, y) that are equidistant to $(3, 5)$ and $(7, -1)$.

61. (a) $y - 5 = \frac{5}{4}(x - \frac{1}{2})$

(b) $(\frac{9}{2}, 10)$

(c) $\overline{PA} = \overline{PB}$

62. 560 ft

63. $(-\frac{7}{2}, -\frac{1}{2})$

64. (a) No

(b) If two triangles have two pairs of congruent sides and the angles formed by the two pairs of sides are congruent, then the two triangles are congruent.

(c) Yes

(d) No

(e) If the angle is acute, there can be two non–congruent triangles that have S.S.A. congruencies

65. –

66. (a) –

(b) It's the set of all points equidistant to $(3, 5)$ and $(7, -1)$.

67. 30 ft

68. 25 ft

69. –

70. (a) –

(b) The bug appears to be moving along a straight line at a constant speed. That assumes the bugs speed and path aren't changing in between the given points

(c) $(6, \frac{19}{2})$

(d) $(\frac{31}{5}, \frac{49}{5})$

71. $(0, \frac{169}{10})$

72. –

73. –

74. (a) $(21, 16)$
    (b) $(30, 22)$
    (c) $(3 + 3t, 4 + 2t)$

75. –

76. –

77. –

78. –

79. Answers may vary.

$$x(t) = 1 + t$$
$$y(t) = 2 + 3t$$

80. (a) The slope between the points is constant at $\frac{2}{3}$.
    (b)
$$\text{slope} = \frac{\text{coefficient on } t \text{ in the } y \text{ equation}}{\text{coefficient on } t \text{ in the } x \text{ equation}}$$
    (c) $y - 5 = \frac{2}{3}(x - 2)$

81. For each journey, $\Delta x = 7$ and $\Delta y = 4$.

82. (a) $[7, 2]$
    (b) $[14, 8]$
    (c) $[-7, -4]$
    (d) $[7, 4]$

83. –

84. (a) 10 units
    (b) 5 units/second
    (c) $[3, 4]$
    (d) $(-3 + 3t, 1 + 4t)$
    (e) $\frac{4}{3}$

85. (a) $[3, 6]$
    (b) $[3, -2]$
    (c) $[-100, 40]$

86. (a) 12 miles east, 16 miles north
    (b) 20 miles
    (c) 10 miles/hour

87. (a) 48 miles
    (b) 4.8 hours
    (c) 28.8 miles east, 38.4 miles north

88. (a) $A'(6,5), B'(8,-2), C'(11,4)$
    (b) Image of B: $(-1,0)$
    Image of C: $(2,6)$

89. On the line: $(7,6), (-3,1), (6,5.5)$
    Not on the line: $(11,7)$

90. –

91. –

92. (a) $[4,-12]$
    (b) $[-4,12]$

93. $(a+2, b-4)$

94. $(0,0), (2,1), (1,2), (3,3)$. Answers may vary.

95. –

96. (a) I would calculate the length of all four sides to verify that there were two separate pairs of congruent sides.
    (b) –
    (c) They aren't congruent.

97. (a) –
    (b) $[-4,-3]$
    (c) -8
    (d) 2 A.M.
    (e) We can determine her position based on the time.
    (f) $(-6, 1.5)$
    (g) $(-4t, 6-3t)$

98. (a)

$$(x(3), y(3)) = (-17, 19)$$
$$(x(2), y(2)) = (-9, 12)$$
$$(x(-2), y(-2)) = (23, -16)$$
$$(x(1.5), y(1.5)) = (-5, 8.5)$$

(b) $y + 2 = -\frac{7}{8}(x - 7)$

(c) –

99. (a) –

(b) 5

(c) $[8, -6]$

100. (a) $(2, 1)$

(b) $[-12, 5]$

(c) –

(d) $(x, y) = (2 - 24t, 1 + 10t)$

101. (a) $[60, -15]$

(b) $[5, 5\sqrt{2}]$

(c) $[\frac{3}{8}, -\frac{1}{3}]$

(d) $[\frac{p^2}{q}, p]$

102. (a) $[-6, 8]$

(b) $[-\frac{3}{5}, \frac{4}{5}]$

(c) $[\frac{3}{5}, -\frac{4}{5}]$

(d) $[6, -8]$

(e) $[\frac{3c}{5}, -\frac{4c}{5}]$

103. (a) No; If two triangles have two pairs of congruent angles and the sides between those two angles are congruent, then the triangles are congruent.

(b) No

(c) –

(d) If two triangles have two pairs of congruent angles and one pair of non-included, corresponding, congruent sides, then the triangles are congruent.

104. –

105. (a) –

(b) $\sqrt{13}$

106. (a) 5

(b) 1,080

(c) 2,016

(d) $5t$

107. –

108. (a)

$$x(t) = 9 + \frac{3}{5}t$$
$$y(t) = -2 + \frac{4}{5}t$$

(b)

$$x(t) = 9 + \frac{3}{20}t$$
$$y(t) = -2 + \frac{4}{20}t$$

(c)

$$x(t) = 9 - \frac{3}{5}t$$
$$y(t) = -2 - \frac{4}{5}t$$

(d)

$$x(t) = 9 - 3t$$
$$y(t) = -2 - 4t$$

109. Converse: If $\angle BAC \cong \angle BCA$ in $\triangle ABC$, then $\overline{AB} \cong \overline{BC}$ and $\triangle ABC$ is isosceles.

110. –

111. (a) Detroit is 132 miles east and 110 miles north of Chicago.
(b) $(x, y) = (3, 5) + t[90, 70]$

112. $(0, -\frac{26}{3}), (\frac{13}{2}, 0)$

113. (a) -2
(b) $\sqrt{5}$
(c) 6
(d) 5 sec
(e) $\sqrt{\frac{29}{5}}$

114. (a) $[-3, 4]$
(b) $[-180, 240]$

(c) $x = -2 - 180t$
$y = 6 + 240t$

115. (a) $t = 1$

(b) A negative t-value denotes a time when the bug hasn't reached the x-axis yet.

116. –

117. (a) She should check if P is on the line perpendicular to $y = 2x - 1$ that runs through A.

(b) No